

TuteDude - PowerBI tutorial

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CREATING DATA MODEL

Topics -

Creating Data Model

Normalization - OLTP and OLAP - reducing redundancy

Facts and Dimensions -

Primary and Foreign Keys

Cardinality

Filter Flow

Common Schemas

Hierarchies

Content -

OLTP - Online Transaction Processing (CRUD operation) - works on normalized data

OLAP - Online Analytical Processing - works on de-normalized data

When multiple tables are not joined - normalization

When multiple tables are joined - de-normalization

Different types of tables -

1. Dimension table - consists of descriptive attributes for grouping and filtering (table having categorical columns)
2. Fact table - provides with numbers which can be associated with groups (table having numerical columns)

NOTE - consider dimension tables as filter tables because they help to filter out values from the fact table.

Primary key - column in table which uniquely identifies each row of that table.

Can't be a null value, duplicate value

Always unique and distinct

Foreign key - columns of a table which are primary key of some other tables.

Foreign key of fact table will relate to primary keys of respective dimension tables

Schema - arrangement of tables

1. Star schema - fact table surrounded by all dimension tables.
2. snowflake schema - extension of star schema which is having dimension tables connected to other dimension tables as well.

Relation between tables -

1. Active - Solid lines; between 2 tables only 1 active relation
2. Inactive - Dotted lines - between multiple inactive relations

Cardinality - kind of relation between tables -

1. one to one : one occurrence of primary key of first table is relating to one occurrence of primary key of second table.
If two tables are exhibiting this relation, it is advised to merge the two tables.
2. one to many : one occurrence of primary key element in first table is having relation with multiple elements of another table.

Always connect primary key of dimension table with foreign key of fact table.

3. many to one : multiple values of a column in first table is connecting to one element of a column in second table.
4. many to many : DON'T NEED THIS RELATION ; happens when one-to-many relation happens from both tables.
Multiple occurrence from first table is having relation with multiple occurrence of second table.
Problem is - Unable to find unique relation between tables.
Solution - create a third table which helps to identify relation between the 2 tables. (this third table will have a primary key)

Filter Context and Filter Flow -

(The arrow which shows in model view between tables)

The filters are applied from dimension table to some column entity present in fact table.

The direction of arrow shows FROM which table filter is coming from and TO which table filter is going to = FILTER DIRECTION

The arrow will always be in one -to-many relation direction.

Bi-directional Filter - Ambiguous nature

While establishing relationship between tables, there is option of cross-filter direction

If this is selected as both = bi-direction relation

Meaning - we can use columns of fact table to filter out elements in the dimension table

Use case - to filter out elements from one fact table to another fact table using dimension table as intermediate.

NOTE - using one fact table to filter out elements of another fact table can miss out some values - ambiguity in performance.

Ambiguity - When more than one bi-directional relation exists between a dimension table and fact tables, there might be ambiguity in the result and thus generates error.

Check the data to see why ambiguity is coming up.

NOTE -

1. prefer to have one-to-many relations
2. prefer to have single direction relations
3. prefer to have data flow from dimension table to fact table

Hide option - to hide certain column from dimension/fact table so that filtering can't be done from them in report view.

Model layout - Helps to look into a portion of the complete layout

Use "Remove from Diagram" option to remove a table from model layout instead of "Delete from Model".

Data Format - The format of the data shown in the PowerBI frontend

Data Type - the format of data saved in the backend of PowerBI

Hierarchy - Tree structure representation of data.

All column of table which exhibits similar relation are put up in hierarchy.

Use case - instead of focusing on different column of table, the analysis can be done on hierarchy in report view.

Calculated Field with DAX - Data Analysis Expression

Topics -

Filtering = Adding or transforming columns

Aggregate functions = creating new insights for table (sum, avg etc.)

New Tables

DAX vs M-Code -

M-Code = Code made by PowerBI at the backend (Power Query). A language to do ETL operations at backend.

DAX = Code in PowerBI at frontend to perform new Measurements.

DAX measures - Aggregate operations applied to a group and provide a single value (eg. sum of order quantity etc.)

Implicit and Explicit Measures for aggregate operations -

Implicit = measures built-in PowerBI

Explicit = measures created by user using DAX

Quick Measure = Templates provided by PowerBI using which explicit measures can be created.

Measures are evaluated based on filter context which means they recalculate whenever the field or filter around them change.

DAX syntax and operators -

[<Name of measurement>] = <Aggregate function>(<'table name>'[column name])

Example - [Total Quantity] = SUM('TRANSACTION TABLE'[QUANTITY])

Here,

Total Quantity -> Measurement Name => Measures are always surrounded by brackets = [Total Quantity]

SUM -> Function Name => Calculated column doesn't require function but Calculated Measure requires function.

'TRANSACTION TABLE' -> Table Name => Table name is surrounded by inverted commas.

[QUANTITY] -> column name => column name is put inside box brackets

NOTE - table name and column name are case-sensitive.

Operators -

1. Arithmetic : +, -, *, /, ^

2. Comparison : >, <, >=, <=, <>, =

3. Text or Logical: & (concatenation), && (logical AND), || (logical OR), IN (logical OR; for an item in given list)

Function Categories -

1. Math and Stats - SUM, AVERAGE, MIN, MAX, DIVIDE, COUNT, COUNTROWS, DISTINCTCOUNT,

These functions are applied as aggregate function i.e. applied to whole column but return single value

Row context functions - ITERATOR, SUMX, AVERAGEX, MAXX, MINX, RANKX, COUNTX

2. Text - Used for columns having textual data

CONCATENATE, COMBINEVALUES, FORMAT, LEFT/MID/RIGHT, UPPER/LOWER, LEN, SEARCH/FIND, REPLACE, SUBSTITUTE, TRIM

3. Date and time -

DATE, DATEDIFF, YEARFRAC, YEAR/MONTH, DAY/HOUR, TODAY/NOW, WEEKDAY, NETWORKDAYS

TIME - DATESYTD, DATESMTD, DATEADD, DATESBETWEEN

4. Logical function - works on multiple columns at a time

IF, IF ERROR, AND, OR, NOT, SWITCH, TRUE, FALSE

5. Table - To create a new table

SUMMARIZE, ADDCOLUMNS, GENERATESERIES, DISTINCT, VALUES, UNION, INTERSECT, TOPN

6. Filter functions -

CALCULATE, FILTER, ALL, ALLEXCEPT, KEEPFILTERS, REMOVEFILTERS, SELECTEDVALUE

7. Relationship functions -

RELATED, RELATEDTABLE, CROSSFILTER, USERRELATION

8. Other categories - TRIGONOMETRY, PARENT/CHILD, INFORMATION FUNCTION

SYNTAX -

SUM -> SUM(<column name>)

AVERAGE -> AVERAGE(<column name>)

MAX -> MAX(<column name>) OR MAX(<scalar list 1>, <scalar list 2>)

MIN -> MIN(<column name>) OR MIN(<scalar list 1>, <scalar list 2>)

DIVIDE -> DIVIDE([<numerator>],[<denominator>], <alternate result: optional>)

COUNT -> COUNT(<column name>) : No Boolean values

COUNTA -> COUNTA(<column name>) : Including Boolean values

DISTINCTCOUNT -> DISTINCTCOUNT(<column name>)

COUNTROWS -> COUNTROWS([<Table name>])

IF -> IF(<logical test>, <result if true>, <result if false>) : returns one value

IFERROR -> IFERROR(<value>,<value if error>) : returns specified value if it returns an error; otherwise the expression is returned.

SWITCH -> SWITCH(<expression>,<value 1>,<result 1>,...,[Else]) : evaluating expression against a list of values and returning one of possible expression.

AND -> AND(<logical 1>,<logical 2>) : evaluates both logical expression and returns a value if both are coming out to be TRUE.

OR -> OR(<logical 1>,<logical 2>) : evaluates both logical expression and returns a value if any of the condition is TRUE.

&& -> works on multiple logical test for AND logical function.

|| -> works on multiple logical test for OR logical functions.

LENGTH -> LEN(<column name>) : extract the number of characters from string

NOTE - <text> also means the name of column

CONCATENATE -> CONCATENATE(<text 1>,<text 2>) : join two strings together into one

UPPER/LOWER -> UPPER(<text>) ; LOWER(<text>) : return Uppercase or Lowercase format data

LEFT/RIGHT/MID -> LEFT(<text>,<number of character>)] ; RIGHT(<text>,<number of characters>)] ; MID(<text>,<start index>,<number of characters>)

SUBSTITUTE -> SUBSTITUTE(<text>,<old text>,<new text>) : replace an instance of text with new text in a string

SEARCH -> SEARCH(<find text>,<within text column>,[<start position>],[<value if not found>]) : returns the position where the specified string or character is found from left to right.

TODAY/NOW -> TODAY() , NOW() : Returns current time and today's date

NOTE - <date> also means the name of column

DAY/MONTH/YEAR -> DAY(<date>) , MONTH(<date>) , YEAR(<date>) : returns day as 1-31, month as 1-12

HOURL/MINUTE/SECOND -> HOUR(<datetime>) , MINUTE(<datetime>) , SECOND(<datetime>) : returns hour as 0-23, minute as 0-59, second as 0-59

WEEKDAY/WEEKNUM -> WEEKDAY(<date>,[<return type>]) , WEEKNUM(<date>,[<return type>])

EOMONTH -> EOMONTH(<start date>,<month value>) : for month value=0 => current month ; for month value=-1 => previous month ; for month value=+1 => next month

DATEDIFF -> DATEDIFF(<first date>,<second date>,<interval>) : returns difference between two dates based on interval

RELATED -> RELATED(<column name>) : Associating a column in dimension table with another column in fact table. Always associate the value in fact table not in dimension table.

CALCULATE -> CALCULATE(<Expression>,[<filter 1>],[filter 2],...) : It's a filter modifier. It's a container for other formulas

ALL -> ALL(<table name or column name>) : returns all of rows in table or column irrespective of all the filters applied. : FILTER REMOVER

It won't return a single value

FILTER -> FILTER(<table name>,<filter expression>) : It goes ROW by ROW. It is related with ITERATOR function.

Filter function parses each and every row and keeps record which satisfies filter expression. It is able to handle complex calculations.

Disadvantage -> it is slow in operation. Slower than CALCULATE.

ITERATOR -> : It's not about the condition but it's about the operation. Iterating through each and every row and some sort of aggregation is performed to result.

FUNCTIONX -> SUMX, COUNTX, MAXX, MINX, AVERAGEX

FUNCTIONX(<table name>,<expression>)

Time Intelligence Patterns - For calculation of common date based comparisons

Performance to Date -> CALCULATE(<measure>,DATESYTD(<date column>))

YTD -> Year to Date

MTD -> Month to Date

QTD -> Quarter to Date

Previous Period -> CALCULATE(<measure>,DATEADD(<date column>,<how much previous in number>,<YEAR/MONTH/QUARTER/DAY>))

Running Total -> CALCULATE(<measure>,DATESINPERIOD(<date column>, MAX(<date column>), <how much previous in number>,<YEAR/MONTH/QUARTER/DAY>))

Visualizing Dashboard

Important questions to ask -

1. What type of Data
2. What needs to be communicated
3. Who is the end user

Types of Data -

1. Time series = conveys information about time (day/month/year etc)
2. Financial = measures like total revenue, total profit etc.
3. Categorical =
4. Geospatial = Information specific to location
5. Textual =
6. Hierarchical
7. Funnel
8. Survey

What needs to be communicated -

1. Comparison
 - Column/Bar chart
 - Clustered Column
 - Data Table/Heat Map

- Radar chart
- Line Chart : used for time series analysis
- Area Chart : used for time series analysis
- 2. Composition - breaking down information among categories
 - Stacked Bar/Column Chart
 - Pie/Donut Chart
 - Stacked Area : Time series
 - Waterfall chart : gains/losses
 - Tree Map/Sunburst : Hierarchical
- 3. Distribution - information distributed between parameters
 - Histogram
 - Density plot
 - Box and Whisper
 - Scatter plot
 - Data table/Heat map
 - Map/Choropleth : geospatial
- 4. Relationship - Correlation between different parameters
 - Scatter plot
 - Bubble chart
 - Data Table/heat map
 - Correlation Matrix

Who is end user

1. Analyst - interested to see details and wants to know what is happening at granular level
 - Tables or combo chars
 - Granular details to support Root cause analysis
2. Manager - wants summarized actionable insights to operate business
 - common charts and graphs
3. Executive - High level information, crystal clear KPIs
 - KPIs cards or simple charts
 - minimal details unless it adds critical context to KPIs

Creating Dashboard in two pattern -

1. F-Pattern - the analysis is centered as letter F
2. Z-Pattern - the analysis is centered as letter Z

Important terms -

Slicers
 HasOne values
 Drill Up and Drill Down
 Drill Through
 Report Interactions
 Bookmarks
 Navigation Button
 Slicer Panels
 Parameters
 Custom tooltip
 Custom visuals
 Mobile Layout

Artificial Intelligence

Anomaly detection (especially in time series data)

AI visuals - Narrative tools, decomposition tree, key influencers, Q&A visuals

Optimizations in PowerBI

Pause Visuals - to stop running queries running in background.

When queries are paused,

1. the report holds all changes and updates, sets report to a visual that has pending state changes.
2. Shows a banner with refresh and resume visual queries.
3. Adds refresh button to individual visuals allowing to refresh that visual
4. Blocks formatting actions.

Refresh Visuals - to refresh the visuals when continuous data streams is coming

Optimization presets - customize the interactive features between different visuals.

Query reduction - doesn't allow interaction among different visuals

Interactivity - interactive nature among different visuals

Customize - user can customize on specific query reduction features.

Apply all slicers button - apply multiple slicers on a report page at once.

Clear all slicers on a report page at once

Apply and clear all slicer buttons, impact all slicers on report page.
There can be multiple buttons like this.
These buttons can be added and formatted just like other buttons.

Performance Analyzer - It records user actions and tracks load time in milli-seconds for each step in process.

It can group actions into buckets.

Three ways in which performance is analyzed -

1. DAX Query - Amount of time it takes for the visual to send the query to backend at power BI engine and then for engine to return the result.
2. Visual Display - Amount of time it takes for visual to show on screen.
3. Other - Amount of time by visual to prepare for query, wait for other visuals to complete their queries.

Power BI new Features and updates -

1. Calculation Group:

Allows to simplify measure management by grouping common calculations into a single table reducing the total number of measures in a model.

Ex, Time intelligence, currency conversion, scenario analysis, dynamic measures etc.

2. New card Visual:

It can help to involve multiple cards into single card so that multiple cards can be grouped together. Scalable Vector graphs or SVGs can be used.

3. DAX Query View:

It's 4th view in powerBI desktop. It allows to use DAX Expressions and keywords to explore, analyze and view data in semantic model.

Build DAX query to return results as a table in powerBI desktop.

Create queries using only one required keyword - Evaluate

Use additional DAX query keywords - ORDER BY, START AT, DEFINE, MEASURE, VAR, TABLE, COLUMN

Use DEFINE to add calculations like Measures, Variables (VAR) and calculated tables and columns.

Explore and add measures using Measure Keyword.

4. Button Slicers:

From Tile slicers to button slicers that helps in putting images to button. Images can be associated with category like employees, faculty etc.

5. Visual Calculations:

These are DAX calculations tied to specific visual on report canvas and are not added to model like traditional columns or measurement. They can't be referenced anywhere else.

There will be an icon associated with measurement as well.

Benefits -

1. no extra columns or measures for storing data in model.
2. visual context supports more intuitive calculations.
3. New DAX templates and functions.

Links -

<https://powerbi.microsoft.com/en-us/blog/>

<https://powerbi.microsoft.com/en-us/blog/category/features/>